

Self-Concept in Consumer Behavior: A Critical Review

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The self-concept literature in consumer behavior can be characterized as fragmented, incoherent, and highly diffuse. This paper critically reviews self-concept theory and research in consumer behavior and provides recommendations for future research.

Most scholars seem to agree that the term "self-concept" denotes the "totality of the individual's thoughts and feelings having reference to himself as an object" (Rosenberg 1979, p. 7). However, self-concept has been treated from various points of view. For example, psychoanalytic theory views the self-concept as a self-system inflicted with conflict. Behavioral theory construes the self as a bundle of conditioned responses. Other views, such as organismic theory, treat the self in functional and developmental terms; phenomenology treats the self in a wholistic form; and cognitive theory represents the self as a conceptual system processing information about the self. Symbolic interactionism, on the other hand, views the self as a function of interpersonal interactions.

Generally, self-concept has been construed from a multidimensional perspective (Burns 1979; Rosenberg 1979). Actual self refers to how a person perceives herself; ideal self refers to how a person would like to perceive herself; and social self refers to how a person presents herself to others. Global self-attitude (e.g., self-esteem or self-satisfaction) has been treated as a conscious judgment regarding the relationship of one's actual self to the ideal or social self (Burns 1979; Rogers 1951).¹

There seems to be a consensus regarding the existence and independent influence of at least two self-concept motives—self-esteem and self-consistency (Epstein 1980). The self-esteem motive refers to the tendency to seek experiences that enhance self-concept. The self-consistency motive denotes the tendency for an individual to behave consistently with her view of herself. Ordinarily, these twin motives are harmonious, but under some circumstances, these same motives conflict (Jones 1973; Schlenker 1975; Shrauger and Lund 1975).²

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PRODUCT SYMBOLISM

In consumer research, Tucker (1957, p. 139) argued that consumers' personalities can be defined through product use:

There has long been an implicit concept that consumers can be defined in terms of either the products they acquire or use, or in terms of the meanings products have for them or their attitudes towards products.

Products, suppliers, and services are assumed to have an image determined not only by the physical characteristics of the object alone, but by a host of other factors, such as packaging, advertising, and price. These images are also formed by other associations, such as stereotypes of the generalized or typical user (cf. Britt 1960; Grubb and Grathwohl 1967; Levy 1959).

Holman (1981) argued that there are at least three con-

¹The structure of the self-concept has been postulated to be characterized along at least nine dimensions—content, direction, intensity, salience, consistency, stability, clarity, verifiability, and accuracy (Rosenberg 1979). Content refers to the inherent aspects of dispositions, social identity elements, or physical characteristics involved in the self-picture. Direction refers to the positivity or negativity of the self-attitude. Intensity refers to the strength of the self-attitude. Salience refers to the extent to which a self-attitude is in the forefront of consciousness. Consistency is the extent to which two or more self-attitudes of the same individual are contradictory. Stability refers to the degree of which a self-attitude does not change over time. Clarity denotes the extent to which a particular self-concept or self-picture is sharp and unambiguous. Verifiability refers to the extent to which a given self-concept is potentially testable or verifiable. Accuracy is the extent to which a given self-concept reflects one's true disposition.

²In addition to this discussion of the self-concept motives, the development of the self-concept was discussed by Rosenberg (1979). He refers to four self-concept formation principles—reflected appraisals, social comparisons, self-attributions, and psychological centrality. Each of these principles guides the development of an individual's self-concept. The reflected appraisal principle refers to the formation of self-concepts based on others' perceptions of oneself. The social comparison principle refers to the influence of one's evaluation of oneself by comparing oneself to significant others. The self-attribution principle refers to the notion that self-concepts are inferred from one's own behavior. And the principle of psychological centrality refers to the hierarchical organization of the self-concepts.

ditions that distinguish products as communication vehicles—visibility in use, variability in use, and personalizability. For a product to have personality associations, it has to be purchased and/or consumed conspicuously or visibly. Variability in use is also important because without variability, no differences among individuals can be inferred on the basis of product use. The personalizability of the product denotes the extent to which the use of a product can be attributed to a stereotypic image of the generalized user. Sirgy (1979, 1980) used the personalizability characteristic as a moderating variable in a self-concept study. Munson and Spivey (1980, 1981) used Katz's (1960) "value-expressiveness" to argue for the effect of product symbolism on the activation of consumer self-concept in consumption-related situations.

At least four different approaches can be identified in self-concept studies that deal directly with product image: (1) product image as it relates to the stereotypic image of the generalized product user; (2) product image in direct association with the self-concept; (3) sex-typed product image; and (4) differentiated product images.

Many self-concept investigators argue that a product image is, in essence, defined as the stereotypic image of the generalized product user, usually measured on a semantic differential scale (e.g., Grubb and Hupp 1968; Grubb and Stern 1971; Schewe and Dillon 1978). Other studies measure product image directly using the semantic differential type of methodology (e.g., Birdwell 1968; Munson and Spivey 1981; Ross 1971; Samli and Sirgy 1981; Sirgy 1979, 1980, 1981a; Sirgy and Danes 1981).

The measurement of the product image in direct association with the self-concept has employed a product-anchored Q-methodology. The respondent is asked to indicate the extent to which a specific product is associated with her actual self-concept, ideal self-concept, and so forth (e.g., Belch and Landon 1977; Greeno, Sommers, and Kernan 1973; Landon 1974; Martin 1973; Sommers 1964).

Sex-typed product image is restricted to those symbolic attributes directly associated with sex roles. This concept has usually been measured using a bipolar masculinity-femininity rating or ranking scale (e.g., Gentry and Doering 1977; Gentry, Doering, and O'Brien 1978; Vitz and Johnston 1965). Other studies, such as Golden, Allison, and Clee (1979) and Allison et al. (1980), have employed two independent constructs to measure masculinity, femininity, and psychological androgyny in product perceptions. Subjects were asked to indicate the extent to which a specific product is masculine on a rating scale ranging from "not at all masculine" to "extremely masculine." The same product was then rated along a similar "femininity" scale. Allison et al. (1980) found that the majority of their respondents perceived masculine and feminine product images as two separate constructs rather than as one dimension (cf. Bem 1974).

Munson and Spivey (1980, 1981) brought out the notion that product images can be activated in various forms. Two possible "product-expressive" self-constructs involve (1) self-perception given a product preference—defined as how one perceives oneself given a preference for a specific prod-

uct, and (2) others' perception of self given a product preference—defined as how a person believes other people view her given a preference for a specific product. However, results showed that consumers may not be able to distinguish between their "own" feelings about a product and their beliefs about how they are viewed by others (cf. Lomander and Spivey 1978).

SELF-CONCEPT IN CONSUMER BEHAVIOR

There is ambiguity and confusion on the precise conceptualization of self-concept in the consumer behavior literature. A number of investigators have discussed self-concept as a single variable and have treated it as the actual self-concept—i.e., as the perception of oneself (e.g., Belenger, Steinberg, and Stanton 1976; Birdwell 1968; Green, Maheshwari, and Rao 1969; Grubb and Hupp 1968; Grubb and Stern 1971). In this vein, self-concept has been labeled "actual self," "real self," "basic self," "extant self," or simply "self." Within the single self-construct tradition, some investigators have restricted self-concept to perceived sex-role (e.g., Gentry and Doering 1977; Gentry, Doering, and O'Brien 1978; Golden et al. 1979).

More recently, Sirgy (1982a, 1982b) has employed the constructs of *self-image value*—the degree of value attached to a specific actual self-concept (a concept parallel to ideal self-concept), and *self-image belief*—the degree of belief or perception strength associated with a self-image (a concept equivalent to the actual self-concept). Furthermore, Schenk and Holman (1980) have argued for the consideration of the *situational self-image*, defined as the result of the individual's repertoire of self-image and the perception of others in a specific situation.

In the multiple self-constructs tradition, self-concept has been conceptualized as having more than one component. Some investigators have argued that self-concept must be treated as having two components—the actual self-concept and the ideal self-concept, defined as the image of oneself as one would like to be (e.g., Belch 1978; Belch and Landon 1977; Delozier 1971; Delozier and Tillman 1972; Dolich, 1969). The ideal self-concept has been referred to as the "ideal self," "idealized image," and "desired self."

Other investigators have gone beyond the duality dimension. Sirgy (1979, 1980) referred to actual self-image, ideal self-image, social self-image, and ideal social self-image. The social self-concept (sometimes referred to as "looking-glass self" or "presenting self") has been defined as the image that one believes others hold, while the ideal social self-concept (sometimes referred to as "desired social self") denotes the image that one would like others to hold (cf. Maheshwari 1974). Hughes and Guerrero (1971) talked about the actual self-concept and the ideal social self-concept. French and Glaschner (1971) used the actual self-concept, the ideal self-concept, and the "perceived reference group image of self" (this latter concept was never formally defined). Dornoff and Tatham (1972) referred to the actual self-concept, ideal self-concept, and "image of

best friend." Sommers (1964) used the actual self-concept and "described other," defined "as if I were this person." Sanchez, O'Brien, and Summers (1975), on the other hand, employed the actual self-concept, ideal self-concept, and the "expected self," which refers to that image somewhere between the actual and the ideal self-concept. Munson and Spivey (1980) referred to the "expressive self," which pertains to either the ideal self-concept or the social self-concept.

Self-Concept Theories

Levy (1959) argued that the consumer is not functionally oriented and that her behavior is significantly affected by the symbols encountered in the identification of goods in the marketplace. His argument, although not regarded as constituting a theory, did serve to sensitize consumer behavior researchers to the potential influence of consumers' self-concepts on consumption behavior.

Following Levy's proposition, a number of self-concept models were formulated to describe, explain, and predict the precise role of consumers' self-concepts in consumer behavior. Rooted in Rogers' (1951) theory of individual self-enhancement, Grubb and Grathwohl (1967) specified that:

1. Self-concept is of value to the individual, and behavior will be directed toward the protection and enhancement of self-concept.
2. The purchase, display, and use of goods communicates symbolic meaning to the individual and to others.
3. The consuming behavior of an individual will be directed toward enhancing self-concept through the consumption of goods as symbols.

Schenk and Holman's (1980) view of situational self-image is based on the symbolic interactionism school of thought. They defined situational self-image as the meaning of self the individual wishes others to have. This situation-specific image includes attitudes, perceptions, and feelings the individual wishes others to associate with her. The choice of which self (actual self, and so on) to express is influenced by the specific characteristics of a given situation. Once an individual decides which image to express in the social situation, she looks for ways of expressing it. The use of products is one means by which an individual can express self-image. Thus, products that are conspicuous, that have a high repurchase rate, or for which differentiated brands are available might be used by consumers to express self-image in a given situation.

The advantages of the concept of situational self-image are that (1) it replaces the proliferating concepts of actual self-image, ideal self-image, and so forth; (2) it includes a behavioral component; and (3) it acknowledges that consumers have many self-concepts. Consumption of a brand may be highly congruent with self-image in one situation and not at all congruent with it in another.

More recently, Sirgy developed a self-image/product-image congruity theory (1981a, 1982a, 1982b). Product cues involving images usually activate a self-schema involving

the same images. For example, a product having an image of "high status" may activate both a self-schema involving the self-concept "I" and a corresponding linkage between that self-concept and the image attribute (self-image) involving "status." This linkage connects the self-concept "I" with the "status" self-image and is referred to as self-image belief. The self-image belief may be either "I am a high status person" or "I am not a high status person." Self-image beliefs are characterized by (1) the degree of belief strength connecting the self-concept "I" with a particular self-image level, and (2) the value intensity associated with the self-image level (e.g., "I like being the high status type").³

Given the activation of a self-schema as a result of a product cue, Sirgy claims that the value placed on the product and its image attributes will be influenced by the evoked self-schema. For instance, if the product is a luxury automobile and its foremost image is a "high status" one, it can be argued that the value inferred for the automobile's "high status" image depends on the precise nature of the evoked self-image dimension involving "status." If "high status" has a positive value on the evoked self-image dimension, then this positive value will be projected to the product; if "high status" has a negative value, then a negative value will be projected to the product image. What is being argued here is that the value or "meaning" of a product image is not independently derived but is, rather, inferred from evoked self-image dimensions.

As Exhibit 1 indicates, a specific value-laden self-image belief interacts with a corresponding value-laden product-image perception, and the result occurs in the form of:

- *Positive self-congruity*—comparison between a positive product-image perception and a positive self-image belief
- *Positive self-incongruity*—comparison between a positive product-image perception and a negative self-image belief
- *Negative self-congruity*—comparison between a negative product-image perception and a negative self-image belief
- *Negative self-incongruity*—comparison between a negative product-image perception and a positive self-image belief.

These different self-image/product-image congruity states will influence purchase motivation differently. Positive self-congruity will determine the strongest level of purchase motivation, followed by positive self-incongruity, negative self-congruity, and negative self-incongruity, respectively. This relationship is explained through the mediation of self-esteem and self-consistency needs.

From a self-esteem perspective, the consumer will be motivated to purchase a positively valued product to maintain a positive self-image (positive self-congruity condition) or to enhance herself by approaching an ideal image (positive self-incongruity condition). The consumer will be motivated to avoid purchasing a negatively valued product

³The strength of the self-image belief parallels the traditional construct of the actual self-concept, whereas the value intensity of the self-image belief seems to be akin to the traditional construct of ideal self-concept (Sirgy, forthcoming).

EXHIBIT 1

THE EFFECTS OF SELF-ESTEEM AND SELF-CONSISTENCY MOTIVES ON PURCHASE MOTIVATION

Self image	&	Product image	result in	Self-image/ product-image congruity	Mediating factors		Purchase motivation
					Self-esteem motivation	Self-consistency motivation	
					leading to		
positive		positive		positive self- congruity	approach	approach	approach purchase motivation
negative		positive		positive self- incongruity	approach	avoidance	conflict
negative		negative		negative self- congruity	avoidance	approach	conflict
positive		negative		negative self- incongruity	avoidance	avoidance	avoidance purchase motivation

to avoid self-abasement (negative self-congruity and self-incongruity conditions). Self-consistency, on the other hand, predicts that the consumer will be motivated to purchase a product with an image (positive or negative) that is congruent with her self-image belief. This functions to maintain consistency between behavior and self-image beliefs (positive and negative self-congruity conditions) and to avoid dissonance generated from behavior/self-image belief discrepancies (positive and negative self-incongruity conditions). The resultant motivational state toward a given product is thus the net effect of the motivational state arising from self-esteem and self-consistency needs.

Self-Concept Measurement

One of the earliest attempts in consumer self-concept measurement was by Sommers (1964). The procedure used was a Q-sort, which groups products on dimensions such as "most like me" to "least like me." Sommers' study provided an initial nomological validation of this procedure.

Many self-concept investigations have employed the Q-sort methodology with relative nomological success (Greeno et al. 1973; Hamm 1967; Hamm and Cundiff 1969; Martin 1973). Belch and Landon (1977) modified the Q-sort by using a rating scale with a predetermined distribution. The methodology was relatively successful in the nomological studies conducted by Landon and his associates (Belch 1978; Belch and Landon 1977; Landon 1972, 1974). A more traditional Q-sort procedure was used in several studies in which personality adjectives were sorted along a self-concept dimension such as "most like me" to "least like me" (French and Glaschner 1971; Sanchez et al. 1975).

Another tradition in self-concept measurement involves the semantic differential. This method entails having the respondent rate a specific self-perspective—actual self-concept, for example—along a number of bipolar adjective

scales (e.g., Bellenger et al. 1976; Birdwell 1968; Delozier 1971; Dolich 1969).

Other miscellaneous measures have also been used to tap the self-concept. These include the adjective check list (Guttman 1973), self-report attitudinal items measured on a Likert-type scale (Jacobson and Kossoff 1963), and other standardized sex-role attitude measures (Gentry and Doering 1977; Gentry et al. 1978; Golden et al. 1979; Morris and Cundiff 1971; Vitz and Johnston 1965).

Self-Concept Research

At least five research tracks directly related to self-concept have been identified:

Self-Concept and Socio-Psychological Factors. Sommers (1964) attempted to differentiate consumers who vary in social stratification (SES) by using self-concept measured in terms of products. A probability sample of 100 housewives and 10 generic products yielded results that were basically consistent with the following hypotheses:

- Members of a high SES stratum (H) describe self significantly differently than do members of a low stratum (L).
- Members of L demonstrate greater agreement in describing self than do members of H.
- Members of H demonstrate greater agreement in describing other consumers than in describing self.
- Members of L demonstrate greater agreement in self description than do members of H.

Martin (1973) and Greeno et al. (1973) attempted to differentiate consumers with varying personalities by using self-concept measured in terms of products. Martin's study employed a nonprobability sample of 223 students, together with two sets of 50 products (one for each sex) from a Sears Catalog. Martin's study revealed three female clusters (per-

sonal hygiene, noncommitted, and liberated) and five male clusters, of which only three were reasonably interpretable (conservative, religious, and personal hygiene). Greeno et al.'s study, which used a probability sample of 190 housewives with 38 generic products, produced six female clusters (homemakers, matriarchs, variety girls, cinderellas, glamour girls, and media-conscious glamour girls). No significant overlap was visible between the female clusters in the two studies, but this could have been due to the different populations (female students versus housewives).

Consumer Behavior as a Function of Self-Concept/Product-Image Congruity. The discussion of actual self-image and product-image congruity was initiated by Gardner and Levy (1955) and Levy (1959). The main attention was focused upon the image projected by various products. Consumers were thought to prefer products with images that were congruent with their self-concepts.

Exhibit 2 includes most of the studies that have examined the relationship between self-concept/product-image congruity and consumer behavior. The findings of these studies can be summarized as follows:

1. The relationship between actual self-image/product-image⁴ congruity (self-congruity) and consumer choice (i.e., product preference, purchase intention, and/or product usage, ownership, or loyalty) has been supported by numerous studies. Those studies which failed to confirm this relationship were Hughes and Guerrero (1971) and Green et al. (1969).
2. The relationship between ideal self-image/product-image congruity (ideal congruity) and consumer choice (i.e., product preference, purchase intention, product usage, ownership or loyalty) has been generally supported.
3. The relationship between social self-image/product-image congruity (social congruity) and consumer choice (limited to product preference, purchase intention, and store loyalty) has not been strongly supported (Maheshwari 1974; Samli and Sirgy 1981; Sirgy 1979, 1980).
4. The relationship between ideal social self-image/product-image congruity (ideal social congruity) and consumer choice (limited to product preference, purchase intention, and store loyalty) has been moderately supported (Maheshwari 1974; Samli and Sirgy 1981; Sirgy 1979, 1980).
5. The relationship between sex-role self-image/sex-typed product-image congruity (sex-role congruity) and consumer choice (limited only to product usage) has been moderately supported (Gentry et al. 1978; Vitz and Johnston 1965).
6. The moderating role of product conspicuousness⁵ on the relationship between self-concept/product-image con-

gruity⁶ and consumer choice (limited to product preference, purchase intention, and/or product usage) has been largely unsupported (Dolich 1969; Ross 1971; Sirgy 1979). That is, it was expected that the ideal and/or ideal-social self-concepts would be more closely related to product preference with respect to highly conspicuous products than to the actual and/or social self-concepts. With respect to inconspicuous products, it was expected that the actual and/or social self-concept would be more closely related to product preference than to the ideal and/or ideal-social self-components.

7. The moderating role of product conspicuousness-social class interaction on the relationship between self-concept/product-image congruity and consumer choice (limited only to product preference) has been suggested by Munson's (1974) study. His results showed that preference for conspicuous products was related to ideal self-concept for upper social class respondents, whereas preference for lower class respondents was not related to either actual or ideal self-concepts for either conspicuous or inconspicuous products.
8. The moderating role of product personalization⁷ on the relationship between self-concept/product-image congruity and consumer choice (limited only to product preference and purchase intention) has been suggested by Sirgy (1979, 1980). That is, the relationship between self-concept/product-image congruity and product preference and purchase intention seems stronger for high personalizing products than for low personalizing products.
9. The moderating role of personality on the relationship between self-concept/product-image congruity and consumer choice (limited to purchase intention) has been suggested by Belch (1978). Belch's results showed that, based on Harvey, Hunt and Schroeder's (1961) personality typology,⁸ System 3 subjects' intentions were more closely related to ideal self-concept than to actual self-concept.

⁴Self-concept is used here in the broad sense, thus denoting any of the self-perspectives, e.g., actual self-concept, ideal self-concept, social self-concept.

⁷Product personalization refers to the extent to which a product has strong image or symbolic associations. Products that are highly personalizing are those which have strong stereotypic images for the general user. This dimension is analogous to the distinction between value-expressive products (high product personalization) and utilitarian-expressive products (low product personalization) made by Locander and Spivey (1978) and Spivey (1977).

⁸Harvey, Hunt, and Schroeder (1961) presented a personality typology based on the notion of cognitive complexity. Four personality types of belief systems were deduced: System 1 persons are those who have a simple cognitive structure and a tendency toward extreme, polarized judgments. They are characterized by high absolutism, closedness of beliefs, high evaluativeness, strong adherence to rules, high ethnocentrism, dogmatism, and authoritarianism. System 2 persons can be described as having somewhat more differentiated and abstract belief systems. They are characterized by an anti-rule and anti-authority orientation. They have low self-esteem and are alienated. System 3 persons are those who have high social needs. System 4 persons represent the most abstract and least constricted of the four belief system. They are characterized by a high task orientation, risk taking, creativity, and relativism; they are more tolerant of ambiguity and flexible in thought and action.

⁴Products as used here are not restricted to tangibles, but apply as well to services, organizations, persons, and so on.

⁵Product conspicuousness is defined as the extent to which a specific product is consumed in public—i.e., the extent of high social visibility or high conspicuousness.

STUDIES RELATING CONSUMER BEHAVIOR WITH SELF-IMAGE/PRODUCT-IMAGE CONGRUITY

^aA number of items in these studies were not reported.

10. The moderating role of personality-product conspicuousness interaction on the relationship between self-concept/product-image congruity and consumer choice (limited to product preference) was suggested by Munson's (1974) dissertation results. Munson used Horney's (1937) personality typology. The results showed that for compliant subjects, preference was somewhat more closely related to actual than to ideal self-concept for inconspicuous products. With respect to both compliant and aggressive subjects, preference was more closely related to the ideal than to actual self-concept for conspicuous products. No clear pattern was revealed with respect to the detached subjects.
11. The moderating role of *type of decision* on the relationship between self-concept/product-image congruity and consumer choice (limited to product preference, purchase intention, and store selection) has been suggested by the findings of Sirgy (1979, 1980) and Dornoff and Tatham (1972). Sirgy's results showed that the ideal and ideal-social self-concepts were more closely related to product preference than to purchase intention, whereas the actual and social self-concepts were more closely related to purchase intention than to product preference. However, this expected finding did not generalize across all products. Dornoff and Tatham found that for routinized decisions (supermarket shopping), actual self-concept was more closely related to store selection than to ideal self-concept and "image of best friend." For non-routine decisions regarding specialty store shopping, "image of best friend" was more closely related to store selection than to actual or ideal self-concepts. With respect to nonroutine decisions regarding department store shopping, store selection was more closely related to ideal self-concept than to actual self-concept or "image of best friend."

Consumer Behavior as a Function of Direct Self-Concept Influences. Those studies which explored this relationship have focused their attention on the effects of self-concept per se rather than on self-concept/product-image congruity. The earliest study in this tradition was conducted by Jacobson and Kossoff (1963), who hypothesized that there is a direct relationship between consumers perceiving themselves as innovative and their attitudes towards small cars. Using a self-concept attitudinal measure of innovativeness and conservatism, and based on a probability sample of 250 respondents, the results showed an *opposite* pattern—i.e., consumers who saw themselves as being conservative were more likely to express a positive attitude than those who saw themselves as innovative.

Guttman (1973) tested the hypothesis that light television viewers perceive themselves as achieving and active, whereas heavy viewers perceive themselves as more sociable. Using 12 personality adjectives in an adjective checklist format, and based on a probability sample of 336 female respondents, the results moderately confirmed the hypothesis.

With respect to the specific effects of sex-role self-concepts, Morris and Cundiff (1971) explored the moderating role of anxiety on product preference of hair spray. Sex-

role self-concept was measured by the femininity scale of the CPI personality inventory on a sample of 223 male students. The results showed an interaction effect between sex-role self-concept and anxiety over preference for hair spray. In the same vein, Gentry and Doering (1977) examined the effects of sex-role self-concept and sex on preference and usage of 10 leisure activities, 13 products and their related brands, and nine magazine types and their related brands. Sex-role self-concept was measured using the femininity scales of the CPI and PAQ personality inventories. Using a sample of 200 students, the results indicated that sex and sex-role self-concept were significant predictors of preference and usage, but the sex variable was the better predictor. Similar findings have been obtained by Golden et al. (1979) and by Allison et al. (1980).

Product Image as a Function of Consumer Behavior. A number of studies in the consumer behavior literature have addressed the relationship between congruity effects and product-image perceptions. Hamm (1967) and Hamm and Cundiff (1969) hypothesized that self-actualization (as measured by the discrepancy between actual and ideal self-images in a product-anchored Q-sort) is related to product-image perceptions. Using a sample of 100 housewives and 50 products, the results provided moderate support to the hypothesis. In the same vein, Landon (1972) hypothesized that need for achievement (as measured by the discrepancy between actual and ideal self-images in a product-anchored Q-methodology) is related to product-image perceptions. Using a sample of 360 students with 12 product categories, the results were found to be consistent with the hypothesis.

In a retail setting and using a sample of 325 female students, Mason and Mayer (1970) found that respondents consistently rated their patronized store as high in status compared to nonpatronized stores. In a study to examine store loyalty determinants, Samli and Sirgy (1981) interviewed 372 respondents in two different stores (a discount store and a specialty clothing store). One of their findings involved high correlations between self-concept/store-image congruity and perceptions and evaluations of functional store-image characteristics. Using a sample of 307 students and 24 products, Golden et al. (1979) and Allison et al. (1980) provided some suggestive evidence concerning the effects of congruence between sex-role self-concept and sex-typed product image on sex-typed product perceptions. Their main finding was an interaction effect between sex-role self-concept, sex, self-esteem, and product type in relation to sex-typed product perceptions.

It should be noted that although these studies argued for a causal type of relationship, they provided correlational data from which causal inferences could not easily be made. Theoretically speaking, this relationship can be explained by what has been referred to in the social psychology literature as "egocentric attribution" and "attributive projection" (Heider 1958; Holmes 1968; Jones and Nisbett 1971; Kelley and Stahelski 1970; Ross, Green, and House 1977). That is, attributing a specific image to a product can be very much affected by the person's *egocentricity*: "I use

it; I am this kind of person; therefore, the product image has to be like me."

Self-Concept as a Function of Behavior Effects. Can consumer behavior affect self-perceptions? This situation can occur when a product image is strongly established and consumers' self-concepts are not articulately formed within a specific frame of reference. For example, a consumer may attribute his usage of a pornographic magazine to his strong need for sexual relations. The formation of the self-image "need for sexual relations" may have been affected by the product image associated with the usage of the pornographic magazine. In social psychology, this phenomenon has been explained by Bem's self-perception theory (1965, 1967).

Indirect evidence for this relationship exists in the consumer self-concept literature. Evans (1968) argued that Birdwell's (1968) study showed that product ownership may have influenced both self-concept and product image, resulting in high self-concept/product-image congruity. The same argument applies to the studies by Grubb and Hupp (1968), Grubb and Stern (1971), and Schewe and Dillon (1978).

In an indirect test of this relationship, Belch and Landon (1977) argued that product ownership influences self-concept measurement (although this was not causally demonstrated). Furthermore, Delozier (1971) and Delozier and Tillman (1972) found that self-concept/product-image congruity increased with the passage of time, which may possibly be indicative of the influence of consumer behavior on self-concept changes.

RESEARCH PROBLEMS

Proliferation of Self-Concept Constructs

Researchers have generated numerous constructs in an attempt to explain consumer self-concept effects on consumer choice. These include ideal self-image, social self-image, expected self-image, situational self-image, and so on. The proliferation of self-concept constructs not only sacrifices theoretical parsimony but also presents theoretical difficulties in describing and explaining the nature of the interrelationship between these constructs. To what extent are these constructs independent of one another? What is the precise nature of their interaction? Under what circumstances? Only recently have some of these issues been addressed.

Schenk and Holman (1980) argued that the situational self-image may offer an integrated and parsimonious approach. The situational self-image is situation-specific and takes into account the actual self-concept, the ideal self-concept, and so on. In the same vein, Sirgy (1981a, 1982a, 1982b, forthcoming) and Sirgy and Danes (1981) argued for the use of self-image/product-image congruity, which takes into account the interrelationship between the self and ideal components of the self-concept, together with product image.

Explanatory Use of Self-Concept Effects

Most self-concept studies to date seem to be based on the congruence notion that consumers are motivated to approach those products which match their self-perceptions, but it is not clear on what theory or theories this congruence notion is based. Rogerian humanistic theory (Rogers 1951) is implicit in the writings of Landon, Grubb, and Ivan Ross. Goffman's (1956) self-presentation theory has been also referenced in a number of studies (e.g., Schenk and Holman 1980; Holman 1981). However, most self-concept studies seem to be atheoretical (e.g., Birdwell 1968; Dolich 1969; Green et al. 1969; Hughes and Naert 1970).

The use of theory is essential in generating testable hypotheses and explaining research findings. Consumer researchers should be encouraged to generate their own self-concept theories in consumption-related settings. In addition, many self-theories in social psychology can be effectively used in consumer research. For example, Festinger's (1954) social comparison theory can be used to explain how consumers evaluate themselves by comparing what they own and consume with others. Bandura's (1977) self-efficacy theory can be employed to explain the difference between ideal congruity and ideal social congruity effects.

Self-concept theories can also be used to guide methodology. Wicklund and Frey's (1980) work on self-awareness can guide methodological attempts to evoke respondents' self-concepts in the research setting. Bem (1967, 1972) cautions us against self-report methods because the inferences made may link respondents' behavior with self-dispositions. Similarly, Deci's (1975) cognitive evaluation theory can be used to explain attributional mechanisms occurring in self-report or survey methodologies. Jourard's (1971) self-disclosure theory explains the biased nature of self-concept reports due to the intimate, personal, and threatening nature of self-concept information.

Self-Image/Product-Image Congruence Models

Modeling self-image/product-image congruity in relation to product preference and purchase intention has been, for the most part, void of theory. Models most predictive of consumer choice or most popular in the research literature have been "automatically" adopted by self-concept researchers.

The mathematical models of self-image/product-image congruity have been examined by a number of investigators in relation to consumer choice. Hughes and Naert (1970) examined the following atheoretical mathematical congruence models in relation to purchase intention:

Simple-difference model	$\sum_{i=1}^n (S_{ij} - P_{ij})$
Weighted simple-difference model	$\sum_{i=1}^n W_{ij} (S_{ij} - P_{ij})$
Simple-difference divisional model	$\sum_{i=1}^n \frac{(S_{ij} - P_{ij})}{P_{ij}}$

Weighted divisional model
$$\sum_{i=1}^n w_{ij} \left(\frac{S_{ij}}{P_{ij}} \right)$$

where

S_{ij} = actual self-image (i) of individual (j)
 P_{ij} = product image (i) of individual (j)
 w_{ij} = importance weight of image (i) of individual (j)

The results showed that weighted simple-difference and weighted divisional models were equally predictive of product choice and more predictive of product choice than the unweighted simple-difference and simple-difference divisional models.

Maheshwari (1974) compared the predictive strength of the Euclidean-distance model $[\sum_{i=1}^n (P_{ij} - S_{ij})^2]^{1/2}$ versus the absolute-difference model $[\sum_{i=1}^n |P_{ij} - S_{ij}|]$ in relation to product preference. The results showed no significant differences between these two congruence models in predicting preference behavior. Sirgy (1981a) and Sirgy and Danes (1981) compared the predictive strength of a model emanating from self-image/product-image congruity theory with the strength of a number of traditionally used congruence models.

Interactive congruence model
$$\sum_{i=1}^n (2P_{ij} - S_{ij})I_{ij}$$

Absolute-difference models
$$\sum_{i=1}^n |P_{ij} - S_{ij}| \text{ and } \sum_{i=1}^n |P_{ij} - I_{ij}|$$

Difference-squared models
$$\sum_{i=1}^n (P_{ij} - S_{ij})^2 \text{ and } \sum_{i=1}^n (P_{ij} - I_{ij})^2$$

Simple-difference models
$$\sum_{i=1}^n (P_{ij} - S_{ij}) \text{ and } \sum_{i=1}^n (P_{ij} - I_{ij})$$

Euclidean-distance models
$$\left(\sum_{i=1}^n (P_{ij} - S_{ij})^2 \right)^{1/2} \text{ and } \left(\sum_{i=1}^n (P_{ij} - I_{ij})^2 \right)^{1/2}$$

Simple-difference-divisional models
$$\sum_{i=1}^n \frac{(P_{ij} - S_{ij})}{S_{ij}} \text{ and } \sum_{i=1}^n \frac{(P_{ij} - I_{ij})}{I_{ij}}$$

where

I_{ij} = ideal self-imagined (i) of individual (j)

The results showed that the interactive congruence model $[\sum_{i=1}^n (2P_{ij} - S_{ij}) I_{ij}]$ was generally equally or slightly more predictive of product preference and purchase intention when compared to the other models.

Some interesting recent developments in communications research have used distance models in multidimensional space as measures of self-concept/product-image congruity

(Woelfel and Danes 1980; Woelfel and Fink 1980). Consumer researchers may benefit from the application of MDS in modeling the congruity process.

Congruence modeling must be guided by theory. Furthermore, any argument for the use of a specific type of cognitive algebra involved in the congruity process should be theoretically positioned in the context of the decision-rule selection and decision-making literatures. Self-concept researchers seem to ignore the work of their colleagues who are decision-making researchers.

Moderator Variables

The use of moderator variables, such as personality differences, social class, and product conspicuousness to moderate the relationship between self-concept/product-image congruity and consumer choice has also been relatively void of theory. For example, Ross (1971) and Dolich (1969) hypothesized that product conspicuousness moderates the relationship between type of self-concept and preference behavior. Specifically, the ideal self-concept was expected to be more closely related to preference for conspicuous products than actual self-concept would be, whereas the actual self-concept was expected to be more closely related to preference for inconspicuous products than ideal self-concept would be. Although this hypothesis sounds plausible, it was not argued within the framework of a particular theory.

A theoretical framework should be selected to hypothesize the moderating effects of particular variables. For example, if we use self-image/product-image congruity theory, it has already been shown that type of consumer decision (attitude toward product versus attitude towards purchase) moderates the effects of self-image/product-image congruity on purchase motivation (Sirgy 1979, 1980, 1982b). Within this theoretical framework, it can be argued that other personality moderator variables (e.g., locus-of-control, self-monitoring, self-esteem, dogmatism, social approval, and achievement motivation) can be used to predict consumer choice. Situational moderator variables may include product conspicuousness, image attainability, purchase conspicuousness, product personalizability, product variability, and perceived risk.

The Semantic Differential

Turning to methodological difficulties, the use of the semantic differential is criticized on many counts. No consensual method is used to select the image adjectives. Some have used general adjectives extracted from personality inventories (e.g., Bellenger et al. 1976; Maheshwari 1974). Others have used attributes most related to the products being tested (e.g., Birdwell 1968; Schewe and Dillon 1978). Only one study (Dolich 1969) used terms that fit Osgood, Succi, and Tannenbaum's (1957) evaluation, potency, activity, stability, novelty, and receptivity factors. It is recommended that the semantic differential methodology only include those images which are most related to the products being tested.

With the exception of Hughes and Naert's work (1970), almost all the studies that employed the semantic differential assumed equal weighting of the image attributes. Since these attributes carry different importance weights for each consumer (Maheshwari 1974), this assumption is clearly unwarranted. It is therefore recommended that importance ratings for each attribute be obtained through self-report methods or other related techniques.

With a few exceptions (Bellenger et al. 1976; Delozier 1971; Delozier and Tillman 1972; Munson 1974; Stern, Bush, and Hair 1977), the majority of studies employing the semantic differential failed to provide evidence of reliability and validity.

Most studies using the semantic differential did not test for attribute interrelationships such as duplication, redundancy, or overlap. Exceptions include Stern et al. (1977), Bellenger et al. (1976), and Maheshwari (1974), who used a factor analytic procedure to reduce the full attribute set. This factor analytic technique is recommended for general use with the semantic differential methodology to ensure attribute independence.

Although one may acknowledge that consumers may see symbolic images in products and that these images interact with their self-images, it can be argued that those images—as tapped by the adjective bipoles in the semantic differential—may not be salient across individuals and across products. Only one or two out of a long list of attributes may be salient in a given consumer's perception of the product and of herself. Thus responses to the nonsalient attributes may present additional methodological confounding. To ensure high image saliency, only those images which are found to be highly related to the product being tested should be included in the semantic differential. In other words, general self-concept standardized scales are *not* recommended.

Further, the semantic differential methodology may be susceptible to halo effects biases. Response to the initial attributes may bias responses on following attributes. Other methodologies free from halo effects could be used to replicate findings from studies using the semantic differential methodology. These other methods may include protocol procedures, free elicitation procedures, and so forth.

It can be argued that the use of bipolar adjectives assumes that consumers can identify with a high degree of certainty which pole of the adjective describes them best. Breaking from this tradition, Grubb and Hupp (1968) and Sirgy (1979, 1980) used unipolar adjectives in a semantic-differential-type format for tapping the degree of applicability or certainty of one's description of oneself along these adjectives. The best possible solution may involve both endorsing an item between the adjectival bipoles and also rating the degree certainty or uncertainty felt regarding item endorsement.

Also, it is not clear how self-concept investigators using the semantic differential methodology avoid social desirability bias (Edwards 1957; Crowne and Marlowe 1964). In an attempt to compensate for social desirability biases in the semantic differential methodology, investigators are advised to (1) select neutral self-image attributes, (2) use both

positive and negative self-image dimensions if that is not feasible, and (3) inform consumers that their responses will remain anonymous (Pryor 1980).

Moreover, the self-image bipolar adjectives used in the semantic differential methodology are very abstract. Bem and Allen (1974) indicated that psychologists measuring self-concept assume that they can measure the relative presence of a particular, abstract self-image characteristic across all persons. However, it is possible that certain abstract self-images may apply to some people but not to others. For example, some consumers may be friendly across a variety of situations. For these consumers, friendliness is a relevant characteristic. Other consumers may be more or less friendly according to the situation: for them, friendliness is not a relevant characteristic. Bem and Allen (1974) recommended at least two approaches to remedy this problem. One possible solution is to make those self-image adjectives situation-specific. This can be accomplished either by instructing consumers to respond to those self-image characterizations while thinking of the product situation being tested, or by phrasing those self-image adjectives in terms of sentence items reflecting a specific consumption situation per self-image, and then using Likert-type scales (instead of the semantic differential scales) in measuring consumers' responses. Another solution involves asking consumers to rate the variation in their self-image characterization across different consumption-related situations.

Finally, image attributes as represented in the semantic differential methodology may create a self-disclosure problem. One central proposition in Jourard's (1971) self-disclosure theory is that generalizations about the self are "intimate" topics that subjects hesitate to disclose. A number of possible solutions are presented that can lessen the confounding effects of the tendency to refrain from self-disclosure. One possible solution is to replace the general personality characterization in the semantic differential methodology with "public self-information" on behaviors. According to the research of Runge and Archer (1979) and Fenigstein, Scheier, and Buss (1975), public self-information on the form of specific behaviors is not perceived to be self-revealing and therefore can lessen the self-disclosure problem.

Another possible solution is to manipulate the immediate environment of the respondents to make it more conducive to self-disclosure. This can be accomplished by (1) placing the respondents in a cozy room with pictures on the wall, cushioned furniture, a rug, and soft lighting (Chaikin, Derlega, and Miller 1976); (2) using an interviewer who may be perceived by the respondents as similar to themselves in many respects (Chaikin and Derlega 1974; Rohrberg and Sousa-Poza 1976); and/or (3) hiring physically attractive interviewers to administer the questionnaire (Brundage, Derlega, and Cash 1977).

The Product-Anchored Q-Method

The product-anchored Q-method is criticized for several shortcomings. For example, some respondents may find it difficult to describe themselves in terms of products (French

and Glaschner 1971). Also, many of the products used do not seem to have strong personality stereotypic associations—e.g., Greeno et al. (1973) used products such as frozen orange juice, shoes, catsup, and potatoes; Belch (1978) and Belch and Landon (1977) used products such as coffee, cameras, and deodorant; and French and Glaschner (1971) used products such as ovens, shoes, refrigerators, and laundry detergent. It is difficult to conceive how these products may have strong personality stereotypic associations, or the extent to which the self-concept may play a role with these sorts of products in determining consumer choice. In addition, the product-anchored Q-method fails to differentiate between product images and self images. This, in turn, prevents attempts to model the self-concept/product-image congruity process. As a result of these irremedial problems, the author does not encourage the utilization of the product-anchored Q-sort in future consumer self-concept investigations.

Standardized Personality Measures

To measure sex-role self-concept, Vitz and Johnston (1965) used the femininity scales of the CPI and MMPI personality inventories. Fry (1971) employed the CPI femininity scale, and Gentry et al. (1978) used those of the CPI and PAQ personality inventories.

It is not clear whether these measures tap *self-perceptions*—what Wylie (1974) calls the “phenomenal self”—or whether they tap hidden, covert, nonconscious personality traits and motives—i.e., the “nonphenomenal self.” Most consumer self-concept investigators seem to assume that self-concept is defined as “the totality of the individual’s thoughts and feelings having reference to himself as an object” (Rosenberg 1979, p. 7). The implicit use of this conceptual definition of self-concept precludes the use of these standardized, “clinical” personality measures as indicators of sex-role self-concept.

Elicitation of Self-Awareness

Wicklund and Frey’s (1980) self-awareness theory postulates that most people focus on the environment because the environment typically provides a high degree of perceptual stimulation, and that self-focused attention is sometimes aversive. Consumer product preference or purchase intention are usually measured in an environment that does not ensure activation of the self-concept. Failing to produce a relationship between the self-concept and product preference or purchase intention can therefore be attributed to the fact that product preference or purchase intention can be determined from a variety of non-self factors. In order to study self-concept influences on these consumer behavior phenomena, a product/situation that will elicit the self-concept must be used.

Pryor (1980) reported on three different methods used to create self-awareness in social psychology studies. One method is sensitizing a person to nuances in his past behavior (i.e., looking back). To induce such “retrospective self-awareness,” social psychologists use videotape feed-

back, diary methods, or instructions eliciting past self-reflections. A second method is to sensitize a person to variations in behaviors as they occur (i.e., self-awareness during behavior). This is usually accomplished through the use of mirrors and/or instructions referring to the self. The third method sensitizes a person to personal characteristics during the process of self-report (i.e., self-awareness during self-report). Again, this is usually done through the use of mirrors and/or specific written or verbal instructions.

CONCLUSION

This paper has attempted to critically review self-concept research. In so doing, various conceptualizations, theories, and models have been discussed and measures used in self-concept studies have been reviewed. Research problems concerning the theoretical and methodological underpinnings of self-concept studies have been identified and recommended solutions have been proposed.

It is disheartening to conclude that, compared to consumer attitude research, consumer self-concept research is in its infancy stage. Much work is needed in theoretical generation, model construction, and method development.

Interest in consumer self-concept research will increase when consumer researchers realize that the knowledge extracted from this type of research is valuable for the applied social science researcher. Such researchers have recently become more comfortable with employing attitude models in applied social research. To date, however, the use of attitude models has been limited to functional attributes and only rarely applied to symbolic or personality-related attributes. Although it would be foolhardy to advocate the use of self-concept/product-image congruity models to the exclusion of the traditional multiattribute attitude models, both types of models should be used to maximize consumer behavior prediction.

Knowledge generated from self-concept research can also contribute to consumer attitude modeling and consumer decision-making research. For some unknown reason, self-concept research has been treated as an offshoot topic that is of interest to some and of little utility to others. Self-concept research is an integral part of attitude research and should be considered as such. Attitude theoreticians and researchers are challenged to develop attitude theories that integrate the social cognitive dynamics involved with both functional and symbolic attributes in explaining, describing, and predicting social behavior.

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REFERENCES

- Allison, Neil K., Linda L. Golden, Gary M. Mullet, and Donna Coogan (1980), “Sex-Typed Product Images: The Effects of Sex, Sex-Role Self-Concept and Measurement Implications,” in *Advances in Consumer Research*, Vol. 7, ed. Jerry Olson, Ann Arbor, MI: Association for Consumer Research, 604–609.

- Bandura, Albert (1977), "Self-Efficacy: Toward a Unifying Theory of Behavioral Change," *Psychological Review*, 84, 191-215.
- Belch, George E. (1978), "Belief Systems and the Differential Role of the Self-Concept," in *Advances in Consumer Research*, Vol. 5, ed. H. Keith Hunt, Ann Arbor, MI: Association for Consumer Research, 320-325.
- and E. Laird Landon, Jr. (1977), "Discriminant Validity of a Product-Anchored Self-Concept Measure," *Journal of Marketing Research*, 14 (May), 252-256.
- Bellenger, Danny N., Earle Steinberg, and Wilbur W. Stanton (1976), "The Congruence of Store Image and Self Image," *Journal of Retailing*, 52 (Spring), 17-32.
- Bem, Daryl J. (1965), "An Experimental Analysis of Self-Persuasion," *Journal of Experimental Social Psychology*, 1, 199-218.
- (1967), "Self-Perception: An Alternative Interpretation of Cognitive Dissonance Phenomena," *Psychological Review*, 74, 182-200.
- (1972), "Self-Perception Theory," in *Advances in Experimental Social Psychology*, Vol. 6, ed. Leon Berkowitz, New York: Academic Press.
- and Andrea Allen (1974), "On Predicting Some of the People Some of the Time: The Search for Cross-situational Consistencies in Behavior," *Psychological Review*, 81 (November), 506-519.
- Bem, Sarah L. (1974), "The Measurement of Psychological Androgyny," *Journal of Consulting and Clinical Psychology*, 42, 155-162.
- Birdwell, Al E. (1968), "A Study of Influence of Image Congruence on Consumer Choice," *Journal of Business*, 41 (January), 76-88.
- Britt, Stewart H. (1960), *The Spenders*, New York: McGraw-Hill.
- Brundage, Lani E., Valerian J. Derlega, and Thomas F. Cash (1977), "The Effects of Physical Attractiveness and Need for Approval on Self-Disclosure," *Personality and Social Psychology Bulletin*, 3 (Winter), 63-66.
- Chaikin, Alan L. and Valerian J. Derlega (1974), "Variables Affecting the Appropriateness of Self-Disclosure," *Journal of Consulting and Clinical Psychology*, 42 (August), 588-593.
- , Valerian J. Derlega, and Sarah J. Miller (1976), "Effects of Room Environment on Self-Disclosure in a Counseling Analogue," *Journal of Counseling Psychology*, 23 (September), 479-481.
- Crowne, W. J. and D. Marlowe (1946), *The Approval Motive: Studies in Evaluative Dependence*, New York: John Wiley.
- Deci, Edward L. (1975), *Intrinsic Motivation*, New York: Plenum, Seligman.
- Delozier, Maynard W. (1971), "A Longitudinal Study of the Relationship Between Self-Image and Brand Image," unpublished Ph.D. thesis, University of North Carolina at Chapel Hill.
- and Rollie Tillman (1972), "Self Image Concepts—Can They Be Used to Design Marketing Programs?" *Southern Journal of Business* 7(1), 9-15.
- Dolich, Ira J. (1969), "Congruence Relationship Between Self-Image and Product Brands," *Journal of Marketing Research*, 6 (February) 80-84.
- Dornoff, R. J. and R. L. Tatham (1972), "Congruence Between Personal Image and Store Image," *Journal of the Market Research Society*, 14, 45-52.
- Edwards, Allen Louis (1957), *The Social Desirability Variable in Personality Assessment and Research*, New York: Dryden.
- Epstein, Seymour (1980), "The Self-Concept: A Review and the Proposal of an Integrated Theory of Personality," *Personality: Basic Issues and Current Research*, ed. Ervin Staub, Englewood Cliffs, NJ: Prentice-Hall.
- Evans, Franklin (1968), "Automobiles and Self Imagery: Comment," *Journal of Business*, 41 (October), 445-459.
- Fenigstein, Allan, Michael F. Scheier, and Arnold H. Buss (1975), "Public and Private Self-Consciousness: Assessment and Theory," *Journal of Consulting and Clinical Psychology*, 43 (August), 522-527.
- Festinger, Leon (1954), "A Theory of Social Comparison," *Human Relations*, 7, 117-140.
- French, Warren A. and Alan B. Glaschner (1971), "Levels of Actualization as Matched Against Life Style Evaluation of Products," *Proceedings of the American Marketing Association*, 30, 358-362.
- Fry, Joseph N. (1971), "Personality Variables and Cigarette Brand Choice," *Journal of Marketing Research*, 8 (August), 298-304.
- Gardner, Burleigh B. and Sidney J. Levy (1955), "The Product and the Brand," *Harvard Business Review*, 33 (April), 33-39.
- Gentry, James W. and Mildred Doering (1977), "Masculinity-Femininity Related to Consumer Choice," *Proceedings of the American Marketing Association Educator's Conference*, 10, 423-427.
- , Mildred Doering, and Terrence V. O'Brien (1978), "Masculinity and Femininity Factors in Product Perception and Self-Image," in *Advances in Consumer Research*, Vol. 5, ed. H. Keith Hunt, Ann Arbor, MI: Association for Consumer Research, 326-332.
- Goffman, Erving (1956), *The Presentation of Self in Everyday Life*, Edinburgh: University of Edinburgh Press.
- Golden, Linda L., Neil Allison, and Mona Clee (1979), "The Role of Sex-Role Self-Concept in Masculine and Feminine Product Perception," *Proceedings of the Association for Consumer Research*, 6, 595-605.
- Green, Paul E., Arun Maheshwari, and Vithala R. Rao (1969), "Self-Concept and Brand Preference: An Empirical Application of Multidimensional Scaling," *Journal of the Market Research Society*, 11(4), 343-360.
- Greeno, Daniel W., Montrose S. Sommers, and Jerome B. Kernan (1973), "Personality and Implicit Behavior Patterns," *Journal of Marketing Research*, 10 (February), 63-69.
- Grubb, Edward L. and Harrison L. Grathwohl (1967), "Consumer Self-Concept, Symbolism, and Market Behavior: A Theoretical Approach," *Journal of Marketing*, 31 (October), 22-27.
- and Gregg Hupp (1968), "Perception of Self, Generalized Stereotypes, and Brand Selection," *Journal of Marketing Research*, 5 (February), 58-63.
- and Bruce L. Stern (1971), "Self-Concept and Significant Others," *Journal of Marketing Research*, 8 (August), 382-385.
- Guttman, Johnathan (1973), "Self-Concepts and Television Viewing Among Women," *Public Opinion Quarterly*, 34 (Fall), 388-397.
- Hamm, B. Curtis (1967), "A Study of the Differences Between Self-Actualization Scores and Product Perceptions Among Female Consumers," *Proceedings of the American Marketing Association*, 26, 275-276.
- and Edward W. Cundiff (1969), "Self-Actualization and Product Perception," *Journal of Marketing Research*, 6 (November), 470-472.

- Harvey, O. J., D. E. Hunt, and H. M. Schroeder (1961), *Conceptual Systems and Personality Organization*, New York: John Wiley.
- Heider, Fritz (1958), *The Psychology of Interpersonal Relations*, New York: John Wiley.
- Holman, Rebecca H. (1981), "Product as Communication: A Fresh Appraisal of a Venerable Topic," in *Review of Marketing*, eds. Ben M. Enis and Kenneth J. Roering, Chicago: American Marketing Association, 106-119.
- Holmes, David S. (1968), "Dimensions of Projection," *Psychological Bulletin*, 69 (April), 248-268.
- Horney, Karen (1937), *The Neurotic Personality of Our Time*, New York: W. W. Norton.
- Hughes, G. David and Phillippe A. Naert (1970), "A Computer-Controlled Experiment in Consumer Behavior," *Journal of Business*, 43 (July), 354-372.
- and Jose L. Guerrero (1971), "Automobile Self-Congruity Models Reexamined," *Journal of Marketing Research*, 8 (February), 125-127.
- Jacobson, Eugene and Jerome Kossoff (1963), "Self-Perception and Consumer Attitudes Toward Small Cars," *Journal of Applied Psychology*, 47(4), 242-245.
- Jones, Edward E. and Richard E. Nisbett (1971), "The Actor and the Observer: Divergent Perceptions of the Causes of Behavior," in *Attribution: Perceiving the Causes of Behavior*, eds. Edward E. Jones et al., Morristown, NJ: General Learning Press.
- Jones, Stephen C. (1973), "Self and Interpersonal Evaluations: Esteem Theories versus Consistency Theories," *Psychological Bulletin*, 79 (March), 185-199.
- Jourard, Sidney M. (1971), *The Transparent Self*, New York: D. Van Nostrand Co.
- Katz, Daniel (1960), "The Functional Approach to the Study of Attitudes," *Public Opinion Quarterly*, 24, 163-204.
- Kelley, Harold H. and Anthony J. Stahelski (1970), "The Social Interaction Bias of Cooperators' and Competitors' Beliefs About Others," *Journal of Personality and Social Psychology*, 16(1), 66-91.
- Lamone, Rudolph P. (1966), "The Use of Semantic Differential in a Study of Self Image, Product Image, and Prediction of Consumer Choice," unpublished Ph.D. thesis, University of Washington, Seattle, WA 98195.
- Landon, E. Laird, Jr. (1972), "Role of Need for Achievement in the Perception of Products," *Proceedings of the American Psychological Association Convention*, 80, 741-742.
- (1974), "Self-Concept, Ideal Self-Concept and Consumer Purchase Intentions," *Journal of Consumer Research*, 1 (September) 44-51.
- Levy, Sidney J. (1959), "Symbols for Sales," *Harvard Business Review*, 37(4), 117-124.
- Locander, W. B. and W. Austin Spivey (1978), "A Functional Approach to Attitude Measurement," *Journal of Marketing Research*, 15, 576-587.
- Maheshwari, Arun K. (1974), *Self-Product Image Congruence: A Macro-Level Analysis*, Ann Arbor, MI: University Microfilms International.
- Martin, W. S. (1973), *Personality and Product Symbolism*, Austin: Bureau of Business, Graduate School of Business, University of Texas.
- Mason, Joseph B., and Morris L. Mayer (1970), "The Problem of the Self-Concept in Store Studies," *Journal of Marketing*, 34 (April) 67-69.
- Morris, George P. and Edward W. Cundiff (1971), "Acceptance by Males of Feminine Products," *Journal of Marketing Research*, 8 (April) 372-374.
- Munson, J. Michael (1974), *Typological Investigation of Self-Concept Congruity and Brand Preferences: Toward a Predictive Model*, Ann Arbor, MI: University Microfilms International.
- and W. Austin Spivey (1980), "Assessing Self-Concept," in *Advances in Consumer Research*, Vol. 7, ed. Jerry Olson, Ann Arbor, MI: Association for Consumer Research, 598-603.
- and W. Austin Spivey (1981), "Product and Brand User Stereotypes Among Social Classes," in *Advances in Consumer Research*, Vol. 8, ed. Kent B. Monroe, Ann Arbor, MI: Association for Consumer Research, 696-701.
- Osgood, Charles E., George J. Succi, and Percy H. Tannenbaum (1957), *The Management of Meaning*, Urbana: University of Illinois Press.
- Pryor, J. B. (1980), "Self-Reports and Behavior," in *The Self in Social Psychology*, eds., Daniel M. Wegner and Robin R. Vallacher, New York: Oxford University Press.
- Rogers, Carl (1951), *Client-Centered Therapy: Its Current Practices, Implications, and Theory*, Boston: Houghton Mifflin.
- Rohrberg, Robert G. and Joaquin F. Sousa-Poza (1976), "Alcohol, Field Dependence, and Dyadic Self-Disclosure," *Psychological Reports*, 39(3), 1151-1161.
- Rosenberg, Morris (1979), *Conceiving the Self*, New York: Basic Books.
- Ross, Ivan (1971), "Self-Concept and Brand Preference," *Journal of Business of the University of Chicago*, 44, 38-50.
- Ross, Lee, David Green, and Pamela House (1977), "The False Consensus Phenomenon: An Attributional Bias in Self-Perception and Social Perception Processes," *Journal of Experimental Social Psychology*, 13 (May), 279-301.
- Runge, J. E. and R. L. Archer (1979), "Reactions to Self-Disclosure of Public and Private Information," unpublished manuscript, University of Texas at Austin.
- Samli, A. Coskun and M. Joseph Sirgy (1981), "A Multi-Dimensional Approach to Analyzing Store Loyalty: A Predictive Model," in *The Changing Marketing Environment: New Theories and Applications*, eds. Ken Bernhardt and Bill Kehoe, Chicago: American Marketing Association, 113-116.
- Sanchez, Humberto T., Terrence V. O'Brien, and George W. Summers (1975), "Self-Concept and Consumer Motivation," *Proceedings of the American Marketing Association Educator's Conference*, 8, 225-227.
- Schenk, Carolyn T. and Rebecca H. Holman (1980), "A Sociological Approach to Brand Choice: The Concept of Situational Self-Image," in *Advances in Consumer Research*, Vol. 7, ed. Jerry Olson, Ann Arbor, MI: Association for Consumer Research, 610-614.
- Schewe, Charles D. and William R. Dillon (1978), "Marketing Information System Utilization: An Application of Self-Concept Theory," *Journal of Business Research*, 6 (January), 67-79.
- Schlenker, Barry R. (1975), "Self-Presentation: Managing the Impression of Consistency When Reality Interferes with Self-Enhancement," *Journal of Personality and Social Psychology*, 32(6), 1030-1037.
- Schrauger, J. Sidney and Adrian K. Lund (1975), "Self Evaluation and Reactions to Evaluations from Others," *Journal of Personality*, 43 (March), 94-108.
- Sirgy, M. Joseph (1979), "Self-Concept in Consumer Behavior," unpublished Ph.D. thesis, Department of Psychology, University of Massachusetts at Amherst.
- (1980), "Self-Concept in Relation to Product Preference and Purchase Intention," in *Developments in Marketing Sci-*

- ence, Vol. 3, ed. V. V. Bellur, Marquette, MI: Academy of Marketing Science, 350–354.
- (1981a), "Testing a Self-Concept Model Using a Tangible Product," in *Proceedings of the American Psychological Association—Consumer Psychology Division*, 89, 17.
- (1981b), "Introducing a Self-Theory to Consumer Personality Research," *JSAS, Catalog of Selected Documents in Psychology*, 11 (May), 33, Ms. 2250.
- (1982a), "Self-Image/Product-Image Congruity and Advertising Strategy," in *Developments in Marketing Science*, Vol. 5, ed. Vinay Kothari, Marquette, MI: Academy of Marketing Science, 129–133.
- (1982b), "Self-Image/Product-Image Congruity and Purchase Motivation: A Role Playing Experiment," *Proceedings of the American Psychological Association—Consumer Psychology Division*, 90.
- (forthcoming), "The Interrelationship Between Self-Congruity and Ideal Congruity in Predicting Purchase Motivation," *Journal of Business Research*.
- and Jeffery Danes (1981), "Self-Image/Product-Image Congruence Models: Testing Selected Mathematical Models," in *Advances in Consumer Research*, Vol. 9, ed. Andrew Mitchell, Ann Arbor, MI: Association for Consumer Research, 556–561.
- Sommers, Montrose S. (1964), "Product Symbolism and the Perception of Social Strata," *Proceedings of the American Marketing Association*, 22, 200–216.
- Spivey, W. Austin (1977), "An Experimental Evaluation of Attitude Change for Attitude Functions Serving Combinations of the Utilitarian and Value-Expressing Attitude Functions," unpublished Ph.D. dissertation, University of Houston, TX.
- Stern, Bruce L., Ronald F. Bush, and Joseph F. Hair, Jr. (1977), "The Self-Image/Store Image Matching Process: An Empirical Test," *Journal of Business*, 50 (January), 63–69.
- Tucker, William Thomas (1957), *Foundations for a Theory of Consumer Behavior*, New York: Holt, Rinehart & Winston.
- Vitz, Paul C. and Donald Johnston (1965), "Masculinity of Smokers and the Masculinity of Cigarette Images," *Journal of Applied Psychology*, 49(3), 155–159.
- Wicklund, R. A. and D. Frey (1980), "Self-Awareness Theory: When the Self Makes a Difference," in *The Self in Social Psychology*, eds. Daniel M. Wegner and Robin R. Vallacher, New York: Oxford University Press.
- Woelfel, J. D. and Jeffrey Danes (1980), "Multidimensional Scaling Models for Communication Research," in *Multivariate Technique in Human Communication of Research*, eds. P. Monge and J. Capella, New York: Academic Press.
- and E. L. Fink (1980), *The Measurement of Communication Processes: Galileo Theory and Method*, New York: Academic Press.
- Wylie, Ruth C. (1974), *The Self-Concept: A Review of Methodological Considerations and Measuring Instruments*, Lincoln, NE: University of Nebraska Press.